Supplement: Auditing Difficulty in Stateful Tool-Use Workflows

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1 Complete Experimental Specification

The primary experiment and follow-on experiment are separately recorded. The primary grid uses $N = 8$, depths $\{1, 2, 4, 8\}$, candidate counts $\{2, 4, 8\}$, active widths $\{1, 2, 4, 8\}$, both snapshot endpoints, and both graph families. Its seeds are 0–255 and each task is evaluated by eight programmed controllers. The follow-on grids use the anchors in the main-text ablation design; their full per-controller configurations are in `results_extended/case_design.json`.

Pairing and randomness. The base generator draws a fixed eight-slot initial array, a permutation of operation identifiers, a shuffled balanced query assignment, update registers, nonzero increments, and candidate permutations. Candidate-count levels filter the same eight-code permutation; active-width levels project query and update indices into the active width. Across work levels, different numbers of operations cause different subsequent random draws, so seed pairing is not exact surface matching. Hidden resampling uses a separate seeded random stream, draws eight fresh values after each success, and replaces only active registers. Failed trials never advance that stream. Controller sampling uses its own stream. Common controller seeds pair variants, but differing controller logic can consume different numbers of draws. These choices are intended for reproducibility, not to certify statistical independence of every repeated condition.

Admission and positive controls. There is no outcome-based rejection or regeneration. All balanced tasks must query every active register; clustered assignments preserve that coverage and its multiplicities. Concentrated-query variants explicitly have effective width one and are tagged as a degenerate manipulation. Positive controls include unrestricted deterministic tracking, residue tracking, deterministic preplanning, unrestricted tracking with a snapshot after every hidden resampling, and candidate elimination with the worst-case error allowance. The runner aborts if a designated positive control fails, if the call horizon is exceeded, or if the adaptive cache exceeds two entries. Analytic probabilities are derived only for supported cases; unsupported predictions are labelled `not_derived`, not replaced with a default or fitted estimate.

Adaptive-cache details. Next-use-2 uses the public deterministic ready schedule to prioritize registers queried in the future. On a visible snapshot, the current query can be answered directly, so future-use selection skips that operation. During history-only feedback, a successful action confirms the queried residue from the chosen candidate code. The controller inserts that residue, applies the public increment if relevant, and evicts values whose next use is furthest away. This is not a generic learned memory strategy; future-query metadata, successful-action feedback, and public updates are deliberate advantages of the control.

Preparation and storage accounting. Residue caches store values modulo b; raw-value caches store values modulo eight. Reported value bits exclude dictionary keys, graph metadata, controller code, temporary operands, and runtime overhead. Preplanning records N candidate slots, using the initial state as preparation workspace; indexed selection records N maps of b code-to-slot entries. The preparation counter charges Nb table entries for each of these representations. It is a workload proxy, not a measurement of exact executed CPU instructions or comparison count in Python's `index` function. Lookup counts are kept separate from predicate comparisons. The experiment does not aggregate these categories into an invented universal compute unit.

2 Calibration Derivations

2.1 Completion states and schedule probabilities

For $k = N/d$ disjoint chains, a completion set is determined by the completed prefix length $0, \ldots, d$ of each chain, giving $(d+1)^k$ reachable masks. Each chain imposes one internal order on its d operations; interleaving the k orders gives $N!/(d!)^k$ valid permutations. A uniformly sampled operation permutation therefore succeeds with probability $(d!)^{-k}$.

For barriers, all previous layers must be complete before any operation in the next layer succeeds. Each of the d layers contributes $2^k - 1$ nonempty partial-completion choices, with one initial empty mask, giving $d(2^k - 1) + 1$ distinct masks. Each layer's operations can be internally permuted in $k!$ ways, so the valid-order count is $(k!)^d$ and random-order success is $(k!)^d/N!$. Failed-action sinks are not included in these completion-mask counts.

2.2 Deterministic history and residue sufficiency

Let C_t be the completed-operation set after t successes. Because each update is additive and public,

$$x_r(t) = x_r(0) + \sum_{v \in C_t : u_v = r} \delta_v \pmod 8.$$

Thus the initial observation and successful-action history determine every register exactly. Repeating snapshots cannot improve optimal information availability in this deterministic protocol, although it can help a restricted retention strategy. Since each tested b divides eight, the same equation modulo b determines every candidate predicate. Tracking $x_r \bmod b$ is therefore sufficient; storing all three raw value bits is not necessary for $b = 2$ or $b = 4$.

A known ready schedule can also be simulated once using the initial observation. Recording its future candidate slots is an exact plan in the deterministic protocol. That construction establishes an alternative representation, not a lower bound on memory and not a claim that planning uses less total storage or preparation.

2.3 Missing-information ceiling

In the hidden-resampling condition, all active values are freshly uniform after each success. Before a masked decision, conditioning on any visible history still leaves the queried code uniform on $0, \ldots, b-1$. Choosing a different ready operation cannot change this fact, because every active register has the same independent fresh distribution. Consequently, the conditional probability of each hidden decision succeeding is at most $1/b$, regardless of additional computation.

With no retries, the number of visible decisions on a successful path is $v = \lceil N/h \rceil$ for snapshot period $h > 0$, or $v = 1$ for initial-only feedback. Applying the chain rule gives $P(\text{complete}) \le b^{-(N-v)}$. An unrestricted exact controller attains this ceiling. The bound does not apply unchanged to recovery protocols: failed trials preserve state and provide information that can support elimination. For scan-limited controllers, formulas that average visible candidate-order draws are ensemble predictions and are labelled separately from exact instance-conditional predictions.

2.4 Finite retry budgets

Uniform replacement guesses have success probability $p = 1/b$ per attempt. If F is the number of errors before N successes, then

$$\Pr(F \le R) = \sum_{r=0}^{R} \binom{N + r - 1}{r} p^N (1-p)^r.$$

Under uniform elimination, the correct code has a uniformly random rank in a permutation of the b candidates. Errors per operation are therefore independent uniform counts on $\{0, \ldots, b-1\}$. The finite-budget probability is

$$\Pr(F \le R) = \frac{1}{b^N} \sum_{r=0}^{R} [z^r](1 + z + \cdots + z^{b-1})^N.$$

The implementation evaluates this coefficient sum by a truncated convolution. If $R \ge N(b-1)$ the probability is one. Mean calls without budget exhaustion are Nb for replacement and $N(b+1)/2$ for elimination. These are exact properties of the selected guessing policies, not general guarantees for arbitrary tool-using controllers.

3 All Follow-On Outcome Summaries

Each row below averages equally over the remaining anchors of its grid. Low and high are descriptive 95% percentile bands from 2,000 paired seed-block bootstrap resamples. They are not simultaneous confidence statements. Constants produce degenerate bands; a sampled zero is not a proof of zero probability. Calls include failed attempts and stop at termination. For exact analytic predictions, progress, error counts, comparison counts, preparation workload, lookups, and cache measurements, see `results_extended/summary.csv`. All 77 paired contrast estimates and their bands are in `results_extended/paired_contrasts.csv`.

Condition notation uses `n` for operation count, `p` for snapshot period, `r` for global error allowance, and `shock1` for independent hidden resampling. Period zero means initial-only state feedback. The `reverse` flag indicates operation identifier reversal. All grids use both graph families.

3.1 Operation Count

Condition	Controller	Success (%)	Low	High	Calls
n16_p0	Cache-2	0.00	0.00	0.00	3.38
n16_p0	Exact	100.00	100.00	100.00	16.00
n16_p0	Random schedule	0.00	0.00	0.00	1.69
n16_p0	Scan-4	50.05	50.00	50.15	9.44
n16_p1	Cache-2	100.00	100.00	100.00	16.00
n16_p1	Exact	100.00	100.00	100.00	16.00
n16_p1	Random schedule	0.00	0.00	0.00	1.69
n16_p1	Scan-4	50.05	50.00	50.15	9.44
n4_p0	Cache-2	13.43	10.94	16.21	2.93
n4_p0	Exact	100.00	100.00	100.00	4.00
n4_p0	Random schedule	10.94	8.11	13.96	1.60
n4_p0	Scan-4	56.59	54.88	58.50	3.13
n4_p1	Cache-2	100.00	100.00	100.00	4.00
n4_p1	Exact	100.00	100.00	100.00	4.00
n4_p1	Random schedule	10.94	8.01	13.96	1.60
n4_p1	Scan-4	56.59	54.88	58.40	3.13
n8_p0	Cache-2	0.73	0.29	1.22	3.17
n8_p0	Exact	100.00	100.00	100.00	8.00
n8_p0	Random schedule	1.17	0.49	1.95	1.63
n8_p0	Scan-4	50.59	50.24	51.03	5.18
n8_p1	Cache-2	100.00	100.00	100.00	8.00
n8_p1	Exact	100.00	100.00	100.00	8.00
n8_p1	Random schedule	1.17	0.49	1.95	1.63
n8_p1	Scan-4	50.59	50.20	51.03	5.18

3.2 Retention Strategy

Condition	Controller	Success (%)	Low	High	Calls
p0	Next-use-2	0.05	0.00	0.15	5.05
p0	Cache-2	0.00	0.00	0.00	2.98
p0	Cache-4	50.00	50.00	50.00	9.66
p0	Residue-8	100.00	100.00	100.00	16.00
p0	Precomputed plan	100.00	100.00	100.00	16.00
p0	Stale-8	0.00	0.00	0.00	4.20
p4	Next-use-2	9.52	7.86	11.28	7.68
p4	Cache-2	0.00	0.00	0.00	3.10
p4	Cache-4	50.00	50.00	50.00	9.76
p4	Residue-8	100.00	100.00	100.00	16.00
p4	Precomputed plan	100.00	100.00	100.00	16.00
p4	Stale-8	2.98	2.05	4.05	5.33

3.3 Snapshot Cadence and Hidden Resampling

Condition	Controller	Success (%)	Low	High	Calls
p0_shock0	Cache-2	0.29	0.05	0.59	2.85
p0_shock0	Exact	100.00	100.00	100.00	8.00
p0_shock0	Scan-4	50.68	50.34	51.12	5.18
p0_shock0	Uniform	0.00	0.00	0.00	1.25
p0_shock1	Cache-2	0.00	0.00	0.00	2.25
p0_shock1	Exact	0.00	0.00	0.00	2.25
p0_shock1	Scan-4	0.00	0.00	0.00	2.01
p0_shock1	Uniform	0.00	0.00	0.00	1.25
p1_shock0	Cache-2	100.00	100.00	100.00	8.00
p1_shock0	Exact	100.00	100.00	100.00	8.00
p1_shock0	Scan-4	50.68	50.34	51.12	5.18
p1_shock0	Uniform	0.00	0.00	0.00	1.25
p1_shock1	Cache-2	100.00	100.00	100.00	8.00
p1_shock1	Exact	100.00	100.00	100.00	8.00
p1_shock1	Scan-4	50.98	50.54	51.46	5.25
p1_shock1	Uniform	0.00	0.00	0.00	1.25
p2_shock0	Cache-2	4.64	3.66	5.71	3.58
p2_shock0	Exact	100.00	100.00	100.00	8.00
p2_shock0	Scan-4	50.68	50.34	51.17	5.18
p2_shock0	Uniform	0.00	0.00	0.00	1.25
p2_shock1	Cache-2	0.29	0.05	0.59	2.49
p2_shock1	Exact	0.29	0.05	0.59	2.49
p2_shock1	Scan-4	0.24	0.05	0.54	2.21
p2_shock1	Uniform	0.00	0.00	0.00	1.25
p4_shock0	Cache-2	0.44	0.20	0.73	2.93
p4_shock0	Exact	100.00	100.00	100.00	8.00
p4_shock0	Scan-4	50.68	50.29	51.07	5.18
p4_shock0	Uniform	0.00	0.00	0.00	1.25
p4_shock1	Cache-2	0.00	0.00	0.00	2.27
p4_shock1	Exact	0.00	0.00	0.00	2.27
p4_shock1	Scan-4	0.00	0.00	0.00	2.03
p4_shock1	Uniform	0.00	0.00	0.00	1.25

3.4 Recovery and Candidate Elimination

Condition	Controller	Success (%)	Low	High	Calls
r0_p0	Cache-2 + eliminate	0.29	0.05	0.59	2.85
r0_p0	Scan-1 + eliminate	0.29	0.10	0.54	1.64
r0_p0	Uniform	0.00	0.00	0.00	1.25
r0_p0	Eliminate	0.00	0.00	0.00	1.25
r0_p1	Cache-2 + eliminate	100.00	100.00	100.00	8.00
r0_p1	Scan-1 + eliminate	0.29	0.10	0.54	1.64
r0_p1	Uniform	0.00	0.00	0.00	1.25
r0_p1	Eliminate	0.00	0.00	0.00	1.25
r2_p0	Cache-2 + eliminate	3.52	2.59	4.49	5.86
r2_p0	Scan-1 + eliminate	2.93	2.10	3.76	5.11
r2_p0	Uniform	0.05	0.00	0.15	3.72
r2_p0	Eliminate	0.00	0.00	0.00	3.92
r2_p1	Cache-2 + eliminate	100.00	100.00	100.00	8.00
r2_p1	Scan-1 + eliminate	2.93	2.10	3.81	5.11
r2_p1	Uniform	0.05	0.00	0.15	3.72
r2_p1	Eliminate	0.00	0.00	0.00	3.92
r56_p0	Cache-2 + eliminate	100.00	100.00	100.00	18.94
r56_p0	Scan-1 + eliminate	100.00	100.00	100.00	21.52
r56_p0	Uniform	78.56	76.61	80.52	43.27

Condition	Controller	Success (%)	Low	High	Calls
r56_p0	Eliminate	100.00	100.00	100.00	27.87
r56_p1	Cache-2 + eliminate	100.00	100.00	100.00	8.00
r56_p1	Scan-1 + eliminate	100.00	100.00	100.00	21.52
r56_p1	Uniform	78.56	76.71	80.47	43.27
r56_p1	Eliminate	100.00	100.00	100.00	27.87
r8_p0	Cache-2 + eliminate	44.63	42.77	46.39	13.63
r8_p0	Scan-1 + eliminate	43.07	41.75	44.38	12.81
r8_p0	Uniform	1.27	0.78	1.81	11.09
r8_p0	Eliminate	6.54	5.42	7.67	12.64
r8_p1	Cache-2 + eliminate	100.00	100.00	100.00	8.00
r8_p1	Scan-1 + eliminate	43.07	41.75	44.43	12.81
r8_p1	Uniform	1.27	0.78	1.81	11.09
r8_p1	Eliminate	6.54	5.42	7.67	12.64

3.5 Query Locality and State Updates

Condition	Controller	Success (%)	Low	High	Calls
balanced_additive	Next-use-2	0.05	0.00	0.15	5.05
balanced_additive	Cache-2	0.00	0.00	0.00	2.98
balanced_additive	Exact	100.00	100.00	100.00	16.00
balanced_additive	Stale-8	0.00	0.00	0.00	4.20
balanced_static	Next-use-2	0.05	0.00	0.15	5.05
balanced_static	Cache-2	0.00	0.00	0.00	3.04
balanced_static	Exact	100.00	100.00	100.00	16.00
balanced_static	Stale-8	100.00	100.00	100.00	16.00
clustered_additive	Next-use-2	1.86	1.32	2.44	7.67
clustered_additive	Cache-2	0.00	0.00	0.00	7.26
clustered_additive	Exact	100.00	100.00	100.00	16.00
clustered_additive	Stale-8	0.15	0.00	0.39	4.80
clustered_static	Next-use-2	1.76	1.17	2.34	7.68
clustered_static	Cache-2	0.00	0.00	0.00	7.24
clustered_static	Exact	100.00	100.00	100.00	16.00
clustered_static	Stale-8	100.00	100.00	100.00	16.00
concentrated_additive	Next-use-2	100.00	100.00	100.00	16.00
concentrated_additive	Cache-2	100.00	100.00	100.00	16.00
concentrated_additive	Exact	100.00	100.00	100.00	16.00
concentrated_additive	Stale-8	9.42	6.93	12.11	7.21
concentrated_static	Next-use-2	100.00	100.00	100.00	16.00
concentrated_static	Cache-2	100.00	100.00	100.00	16.00
concentrated_static	Exact	100.00	100.00	100.00	16.00
concentrated_static	Stale-8	100.00	100.00	100.00	16.00

3.6 Candidate Selection and Resource Controls

Condition	Controller	Success (%)	Low	High	Calls
b2	Exact	100.00	100.00	100.00	8.00
b2	Indexed	100.00	100.00	100.00	8.00
b2	Scan-1	100.00	100.00	100.00	8.00
b2	Sham-check-4	100.00	100.00	100.00	8.00
b2	Scan-2	100.00	100.00	100.00	8.00
b2	Fixed-check-4	100.00	100.00	100.00	8.00
b4	Exact	100.00	100.00	100.00	8.00
b4	Indexed	100.00	100.00	100.00	8.00
b4	Scan-1	0.59	0.00	1.37	2.00
b4	Sham-check-4	0.59	0.00	1.37	2.00

Condition	Controller	Success (%)	Low	High	Calls
b4	Scan-2	9.18	6.64	12.11	3.55
b4	Fixed-check-4	100.00	100.00	100.00	8.00
b8	Exact	100.00	100.00	100.00	8.00
b8	Indexed	100.00	100.00	100.00	8.00
b8	Scan-1	0.00	0.00	0.00	1.25
b8	Sham-check-4	0.00	0.00	0.00	1.25
b8	Scan-2	0.00	0.00	0.00	1.61
b8	Fixed-check-4	1.17	0.20	2.34	2.39

3.7 Operation Identifiers

Condition	Controller	Success (%)	Low	High	Calls
reverse0_p0	Cache-2	0.29	0.05	0.59	2.85
reverse0_p0	Exact	100.00	100.00	100.00	8.00
reverse0_p0	Random schedule	0.00	0.00	0.00	1.29
reverse0_p0	Scan-4	50.68	50.34	51.12	5.18
reverse0_p1	Cache-2	100.00	100.00	100.00	8.00
reverse0_p1	Exact	100.00	100.00	100.00	8.00
reverse0_p1	Random schedule	0.00	0.00	0.00	1.29
reverse0_p1	Scan-4	50.68	50.29	51.12	5.18
reverse1_p0	Cache-2	0.15	0.00	0.34	2.88
reverse1_p0	Exact	100.00	100.00	100.00	8.00
reverse1_p0	Random schedule	0.00	0.00	0.00	1.35
reverse1_p0	Scan-4	50.44	50.15	50.83	5.23
reverse1_p1	Cache-2	100.00	100.00	100.00	8.00
reverse1_p1	Exact	100.00	100.00	100.00	8.00
reverse1_p1	Random schedule	0.00	0.00	0.00	1.35
reverse1_p1	Scan-4	50.44	50.10	50.83	5.23

4 All Endpoint Interaction Estimates

The following estimates are exploratory follow-on analyses of the original 192-cell factorial experiment. D, B, W, O denote depth, candidate count, active width, and history-only versus snapshot feedback. Endpoint contrasts use high minus low for each participating dimension and average all nonparticipating controls over their original levels. A pairwise interaction is a difference-in-differences; the corresponding signed construction generalizes to orders three and four. Values are percentage points, so an interaction is not itself a success probability. Per-estimate bootstrap bands are retained in `results_extended/all_endpoint_interactions.csv`.

Family	Controller	$D \times B$	$D \times W$	$D \times O$	$B \times W$	$B \times O$	$W \times O$
Barriers	exact	0.00	0.00	0.00	0.00	0.00	0.00
Barriers	memory2	-0.10	0.10	0.10	-2.05	-2.05	-97.95
Barriers	memory2_scan4	0.49	-1.27	0.10	48.93	47.56	-50.68
Barriers	memory4	0.68	-0.68	-0.68	-4.00	-4.00	-95.61
Barriers	random_order	0.00	0.00	0.00	0.00	0.00	0.00
Barriers	scan1	0.00	0.00	0.00	0.00	0.00	0.00
Barriers	scan4	-0.59	0.59	0.00	2.93	0.00	0.00
Barriers	uniform	-0.59	0.98	0.00	-0.59	0.00	0.00
Chains	exact	0.00	0.00	0.00	0.00	0.00	0.00
Chains	memory2	-0.10	0.10	0.10	-2.05	-2.05	-97.95
Chains	memory2_scan4	0.49	-1.27	0.10	48.93	47.56	-50.68
Chains	memory4	0.68	-0.68	-0.68	-4.00	-4.00	-95.61
Chains	random_order	0.00	0.00	0.00	0.00	0.00	0.00
Chains	scan1	0.00	0.00	0.00	0.00	0.00	0.00
Chains	scan4	-0.59	0.59	0.00	2.93	0.00	0.00
Chains	uniform	-0.59	0.98	0.00	-0.59	0.00	0.00

Family	Controller	$D \times B \times W$	$D \times B \times O$	$D \times W \times O$	$B \times W \times O$	$D \times B \times W \times O$
Barriers	exact	0.00	0.00	0.00	0.00	0.00
Barriers	memory2	-0.20	-0.20	0.20	-4.10	-0.39
Barriers	memory2_scan4	-1.37	1.37	0.20	95.12	2.73
Barriers	memory4	1.37	1.37	-1.37	-8.01	2.73
Barriers	random_order	0.00	0.00	0.00	0.00	0.00
Barriers	scan1	0.00	0.00	0.00	0.00	0.00
Barriers	scan4	1.17	0.00	0.00	0.00	0.00
Barriers	uniform	-1.95	0.00	0.00	0.00	0.00
Chains	exact	0.00	0.00	0.00	0.00	0.00
Chains	memory2	-0.20	-0.20	0.20	-4.10	-0.39
Chains	memory2_scan4	-1.37	1.37	0.20	95.12	2.73
Chains	memory4	1.37	1.37	-1.37	-8.01	2.73
Chains	random_order	0.00	0.00	0.00	0.00	0.00
Chains	scan1	0.00	0.00	0.00	0.00	0.00
Chains	scan4	1.17	0.00	0.00	0.00	0.00
Chains	uniform	-1.95	0.00	0.00	0.00	0.00

5 Workshop Comparison and Submission Scope

We inspected four ALA 2025 papers as a purposive sample, not all workshop papers and not a systematic novelty search. Their total PDF lengths are nine pages (failure sampling), ten (Dynamic Thinker), ten (task-graph scheduling), and eight (deliberation cost). References begin on pages nine, eight, seven, and eight, respectively. Total PDF length therefore does not equal the allowed main-text length.

The historical ALA 2025 call allows original work up to eight pages excluding references, and preliminary or outlook papers up to six pages. It specifies double-blind review and deanonymized camera-ready copies. Our expanded authored version is a preprint-style manuscript with an appendix; the separate `review.tex` source removes the author block and appendix, but must still be checked against the current workshop's track and page rules. A longer document is not evidence of acceptance or of a stronger contribution. No current-edition acceptance, conference metadata, or submission date is asserted.

6 Reproduction and Evidence Boundaries

The benchmark, controllers, tests, and analysis use the Python standard library. Document inspection additionally uses the pinned PDF packages. From the project directory in a Windows PowerShell terminal:

```
.\.venv\Scripts\python.exe -m unittest -v test_benchmark test_extended
.\.venv\Scripts\python.exe run_experiments.py --seeds 256
.\.venv\Scripts\python.exe analyze.py
.\.venv\Scripts\python.exe run_extended.py
.\.venv\Scripts\python.exe analyze_extended.py
.\build.ps1
.\.venv\Scripts\python.exe verify_artifact.py
.\.venv\Scripts\python.exe package_artifact.py
```

Re-running the primary experiment replaces its generated records; the source and protocol hashes must match the supplied metadata when reproducing that study. The original and follow-on episode files have separate hashes and execution metadata. The two locally frozen protocols and the time ordering of exploratory analyses are retained explicitly. Complete primary contrasts, rare-scheduler Wilson intervals, and example traces remain available in the original result directory.

This artifact contains only programmed-controller evidence. No frontier model, learned policy, training run, provider reasoning-budget intervention, human participant, or realistic transfer task has been evaluated. Source compilation and local reproducibility do not establish arXiv moderation approval, workshop acceptance, or compatibility with a particular submission processor. The author must review the manuscript and current submission rules before uploading it.